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Information Theoretic Analysis of Questionnaire Data

PATRICK L. BROCKETT, PERRY D. HAALAND, AND ARNOLD LEVINE

Abstract—Two important problems in the analysis of categorical questionnaire data are considered: assessment of question worth and variable selection, and discrete discriminant analysis when the data is nonordinal with many states and few respondents. The unifying approach used throughout is the concept of information theoretic distance measures. Simulations and applications to real data are presented.

I. INTRODUCTION

IN QUESTIONNAIRE analysis, a question is called "closed" if the respondent must select his response from a list of permitted answers. Multiple choice questions are examples of closed questions. If the answers are cate-

gories or qualities, the question is also called a categorical question, and a questionnaire consisting of a series of categorical questions is called a categorical questionnaire. The type of categorical questionnaire we shall discuss in this paper is one in which the respondent is instructed to answer every question. (This is in contrast with the "branching" or "tree" or "funneling" type of questionnaire where such instructions appear as "If your answer to question 1 is a, go to question 5; if your answer is b, go to question 6.") Branching types of questionnaires can be treated by graph theory (Picard [1]).

The analysis of categorical questionnaires poses many interesting problems of which we shall consider two: the assessment of which questions are worthwhile and which questions should be excluded (variable selection), and the problem of discriminant analysis. These two problems are considered here as variants of a single problem which we shall attack using information theoretic techniques.

The use of information theoretic techniques is especially appealing in the analysis of questionnaire data since the entire purpose of such data is to answer some specific queries, and the worth of each question should be de-

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terminated according to how much information is supplied by the question toward answering these queries. To make this mathematically rigorous, suppose we wish to decide whether a respondent belongs in group 1 or group 2, with respective generalized densities f_1 and f_2 with respect to some measure λ . If the prior probabilities of group i membership are π_i , $i = 1, 2$, then the log odds ratio in favor of group 1 membership is $\ln \pi_1/\pi_2$. If an observation x is made on the respondent, Bayes' theorem may be used to determine the new posterior log odd ratio in favor of group 1 membership. The difference between the posterior and prior log odds ratio is taken as a measure of the amount of information supplied by the observation x for discrimination in favor of group 1 membership. One easily works out that this difference is $\ln(f_1(x)/f_2(x))$, and this quantity is called the information gain from the observation in favor of group 1 membership, or simply the information gain (cf. Kullback [2]). The expected information gain is obtained by randomizing x according to the density f_1 obtaining the directed information measure

$$I(f_1|f_2) = \int \ln \frac{f_1(x)}{f_2(x)} f_1(x) \lambda(dx).$$

The symmetric measure of information between the two groups is called the divergence between the groups and is denoted by

$$\begin{aligned} J(f_1, f_2) &= I(f_1|f_2) + I(f_2|f_1) \\ &= \int (f_1(x) - f_2(x)) \ln \frac{f_1(x)}{f_2(x)} \lambda(dx). \end{aligned}$$

For the categorical questionnaires we shall be considering, we take λ as counting measure and the integrals become summations: $J(f_1, f_2) = \sum_i (p_i - q_i) \ln(p_i/q_i)$ where $f_1(x_i) = P_1[X = x_i] = p_i$ and $f_2(x_i) = P_2[X = x_i] = q_i$.

It should be noted that information theoretic methods have been proposed before for branching, or funneling type questionnaires. In these type of questionnaires, information coding and transmission ideas can be used successfully for analysis. This approach is quite different from the one presented here for analysis of nonbranching categorical questionnaires which have an essentially multinomial structure (as opposed to graph theoretic structure). In these previous studies discussion on sequencing of questions occur, while here we assume a fixed questionnaire structure is presented to all respondents. For further discussion of information theoretic techniques in branching questionnaires, see Terrenoire [3] and Picard [1]. Our work is different from others in that we emphasize the "distance measure" interpretation of the divergence and the statistical distribution properties of the sample estimates. Previous uses of information theory in questionnaire analysis emphasized coding and stochastic sequencing of questionnaires. For a review of these methods applied to convergence of "pseudoquestionnaires" see Terrenoire [3]–[5], and Chenais and Terrenoire [6].

II. VARIABLE SELECTION: ASSESSING QUESTION "WORTH"

One purpose of a questionnaire may be to obtain a summary index of how much of a certain attribute is possessed by the respondent. For example a psychiatric screening exam might measure how much "mental stress" is exhibited by a respondent, while in an industrial context, a quality control checklist might measure how much "propensity to fail" is exhibited by a certain machine. Employment screening exams which hope to measure a candidate's potential job success are another example. In each case a value is assigned in some manner to each possible response to a question. The overall questionnaire score of a respondent is obtained by summing the values of his responses.

One method of assessing the discriminatory power of a particular question is to compare a respondent's overall score with that obtained on that particular question. The common approach described by Goode and Hatt [7] is as follows:

In calculating discriminative power the investigator sums the arbitrarily scored items for each person and places the sheets or scores in an array (from the lowest to the highest scores). He then selects out those who fall (in the upper quartile) and who fall (in the lower quartile) ... the upper and lower quartiles are compared.

The method we propose here is in this vein. We divide the respondents into quartiles, Q_1 , Q_2 , Q_3 , and Q_4 , based upon their overall questionnaire scores excluding the question we wish to assess, and we measure the worth of that particular question by the amount of information it possesses for discriminating between the high and low scorers. (Q_1 and Q_4 respondents, respectively)

Let $\hat{p}_1, \hat{p}_2, \dots, \hat{p}_k$ denote the proportion of high scorers (group Q_4) and $\hat{q}_1, \hat{q}_2, \dots, \hat{q}_k$ denote the proportion of low scorers (Q_1) responding to the k permitted answer to the question under consideration. Suppose there are n respondents in each of the reference groups Q_1 and Q_4 (e.g., $\hat{p}_i = (\text{number in group 1 responding } i)/n$). A measure of the amount of information in the question for discriminating between Q_1 and Q_4 (and hence a measure of the worth of the question) is given by taking a linear function of the estimated information theoretic divergence between Q_1 and Q_4 on the question. We define the D -value of the question to be

$$D = n/2 \sum_{i=1}^k (\hat{p}_i - \hat{q}_i) \ln(\hat{p}_i/\hat{q}_i).$$

We would discard a question if D is too close to zero, indicating that there is not sufficient information furnished by the question to discriminate between the high and low questionnaire scorers. The philosophy here is that any question that cannot discriminate between the extremes is of little value. Kullback [2] shows that under a null hypothesis of $(p_1, \dots, p_k) = (q_1, \dots, q_k)$ (corresponding to the question having no discriminatory value), the asymptotic distribution of D is $\chi^2(k-1)$. Thus our procedure is to retain a question only if $D > \chi^2_{1-\alpha}(k-1)$ where $\chi^2_{1-\alpha}(k$

— 1) is the $1 - \alpha$ th quantile of the $\chi^2(k - 1)$ distribution, and α is a prechosen threshold level. The probability of erroneously including a nondiscriminating question by using this procedure converges to α as the sample size increased. For questionnaires such as employment screening questionnaires in which, for legal reasons, each question's inclusion must be justified, this property is very desirable. One can then conclude at a predetermined level of significance that each question included does indeed discriminate.

The above approach can also be used for assessing the discriminatory power of a question for distinguishing between two *pre-established* groups. Assume we have n_1 respondents from group 1 and n_2 respondents from group 2. For a particular question X , let $\hat{p}_X$ and $\hat{q}_X$ represent the empirical response probabilities of groups one and two, respectively, over the k_X categorical answers. The empirical divergence for question X is

$$D(X) = \frac{n_1 n_2}{n_1 + n_2} \sum_{x=1}^k (\hat{p}_x - \hat{q}_x) \ln(\hat{p}_x / \hat{q}_x). \quad (1)$$

It tells us the amount of information available in question X for discriminating between the two known groups. Asymptotically $D(X)$ is $\chi^2(k_X - 1)$ for nondiscriminating questions, whereas for discriminating questions it is noncentral $\chi^2(k_X - 1)$ with noncentrality parameter equal to the discriminatory power of question (cf. Kullback [2]). Again, only questions with $D(X)$ significantly greater than zero are retained.

III. A SEQUENTIAL VARIABLE SELECTION TECHNIQUE

The previous variable (question) assessing scheme can be extended to obtain a sequential selection technique. If each individual question's worth need not be established, however, the overall questionnaire is to contain as much information as possible, then the following sequential method may be used. The general philosophy we use is to select questions in order of decreasing *additional* information furnished for discriminating between the groups. When the amount of additional information furnished falls below some given level γ , the selection scheme terminates. Again we let $\hat{p}_X$ and $\hat{q}_X$ denote the observed response probabilities for the n_1 respondents from group 1 and n_2 respondents from group 2, respectively, over the k_X categorical answers for question X .

Our sequential procedure begins by choosing for first inclusion the most informative question X for discriminating. In this first step our procedure is similar in philosophy to that described by Levine [8], Brockett, Haaland, and Levine [9], and by Goldstein and Dillon [10] (see also Goldstein and Dillon [11], for selecting binary variables for inclusion in a multiway contingency table discrimination framework). In our case we do not assume that two categorical questions have the same number permissible categorical responses, e.g., the questions "sex" and "income level" may have a markedly different number of response categories. This prohibits us from using the Goldstein–

Dillon procedure. We shall use the quantity $D(X)$ from (1) as a measure of information in question X for discrimination (cf. Kullback [2], Gokhale and Kullback [12]). Asymptotically $D(X)$ has a $\chi^2(k_X - 1)$ distribution, and this is why direct comparison of the calculated $D(X)$ values is impossible. A question with $k_X = 11$ answers is expected to have a $D(X)$ value of 10, whereas a question with $k_X = 2$ would be expected to have a $D(X)$ value of 1 under the hypothesis that the question is not discriminating. This does not directly imply, however, that the question with 11 answers is more informative than the question with two answers.

Since $D(X)$ has a $\chi^2(k_X - 1)$ distribution under the null hypothesis that the two groups behave the same on that question, and since it has a noncentral $\chi^2(k_X - 1)$ distribution with a noncentrality parameter equal to the discriminatory power of the question in the case where the alternative hypothesis holds and the question actually discriminates, we shall use the p -value of the $D(X)$ statistic as a measure of the discriminatory power of question X . If $p_X = P[\chi^2(k_X - 1) \geq d]$ where d is the observed value for $D(X)$, then the smaller p_X , the more informative is question X . Although the values $D(X)$ for various different X may not be directly comparable in general (as they would be, for example, if k_X was always the same), the p -values p_X are always comparable and easily calculated from readily available χ^2 tables. If no χ^2 tables are available, one may easily calculate the required p_X values via the probability integral transform. For $v = 1$ degree of freedom we have $P[\chi_{(1)}^2 \geq d] = 2(1 - \Phi(\sqrt{d}))$; for $v = 2$ degrees of freedom we have $P[\chi_{(2)}^2 \geq d] = e^{-d/2}$, while for $v \geq 3$ degrees of freedom we may use the fact that $(\chi_{(v)}^2/v)^{1/3}$ is approximately $N(1 - 2v/9, 2v/9)$ to obtain

$$\begin{aligned} P[\chi_{(v)}^2 \geq d] &= P[(\chi_{(v)}^2/v)^{1/3} \geq (d/v)^{1/3}] \\ &= 1 - \Phi\left(\frac{9(d/v)^{1/3} - 9 + 2v}{\sqrt{18v}}\right), \end{aligned}$$

where Φ is the cumulative distribution function (cdf) for the standard $N(0, 1)$ variate. These formulas make the calculation of the p_X values quite simple, and the approximations involved are quite good (see Zar [13]).

Using the p -values, which are distributed uniformly over $[0, 1]$ under the null hypothesis of no discriminatory power, we select as the first question that question X with minimum p_X value, provided this p_X value is significantly small. We can assess significance for $\min_{1 \leq X \leq m} p_X = U_{(1)}$ by using the distribution function $F(t) = 1 - (1 - t)^m$ for $0 \leq t \leq 1$ as the cdf for $\min_{1 \leq X \leq m} p_X$. Thus the best question has significant discriminatory power at level of significance α if $\min_{1 \leq X \leq m} p_X \leq 1 - (1 - \alpha)^{1/m}$. (The Goldstein–Dillon procedure does not employ the actual distribution for their selection statistic, and hence will not lead to a fixed type 1 error.) Having chosen the first question for inclusion according to this procedure, we select the second question for inclusion as that question which yields the maximum *additional information* to the already selected first question.

For notational convenience, relabel the questions so that the first question selected is called question 1. We look at all question pairs $(1, Y)$, $2 \leq Y \leq m$ and consider the joint probabilities P_{xy} and q_{xy} for the two groups over the possible answer pairs (x, y) on question $(1, Y)$.

The quantity

$$D(1, Y) = n_1 n_2 / (n_1 + n_2) \sum_{x=1}^{k_1} \sum_{y=1}^{k_Y} (p_{xy} - q_{xy}) \ln p_{xy} / q_{xy}$$

is a measure of the joint discriminatory power of the question pair $(1, Y)$, and hence $D(1, Y) - D(1)$ is a measure of the increase in discriminatory information obtained by adding question Y . Note that

$$\begin{aligned} D(1, Y) - D(1) &= n_1 n_2 / (n_1 + n_2) \sum_x \sum_y (p_{xy} - q_{xy}) \ln (p_{xy} / q_{xy}) \\ &\quad - n_1 n_2 / (n_1 + n_2) \sum_x (p_x - q_x) \ln (p_x / q_x) \\ &= n_1 n_2 / (n_1 + n_2) \sum_x \sum_y (p_{xy} - q_{xy}) \ln (p_{xy} q_x / p_x q_{xy}) \\ &= n_1 n_2 / (n_1 + n_2) \sum_x p_x I(p_{Y|x} | q_{Y|x}) \\ &\quad + n_1 n_2 / (n_1 + n_2) \sum_x q_x I(q_{Y|x} | p_{Y|x}) \end{aligned}$$

where $p_{Y|x}$ and $q_{Y|x}$ are the conditional probability distributions of groups 1 and 2, respectively, over the answers to question Y given that question 1 was answered x , i.e., $p_{Y|x}(y) = p_{xy} / p_x$. This equality implies $D(1, Y) - D(1) \geq 0$ with equality only if Y contains no additional information given the answer to question 1 (i.e., the addition of Y can only increase discrimination). This equation also shows that the distribution of $D(1, Y) - D(1)$ is a weighted sum of (nonindependent) χ^2 variables, the weights reflecting the probability of a particular response x to question 1, and the χ^2 variable measuring the information expected to be added by question Y given that particular response x to question 1. A stepwise regression analog would consider $\{D(1, Y) - D(1)\} / D(1)$ as a measure of increased discriminatory power obtained by the addition of question Y . When the question responses are independent variates, one has $D(1, Y) = D(1) + D(Y)$ so the above ratio has (asymptotically) an F distribution with parameters $(k_Y - 1, k_1 - 1)$ (see Brockett, Haaland, and Levine [14]).

In this paper we shall also present a different approach based upon information theoretic analysis similar to that used in contingency table analysis (cf. Gokhale and Kullback [12] and Kullback [2]). This method is more convenient to use than Goldstein and Dillon's technique since one utilizes the entire set of answers to the previous question for determining the usefulness of a proposed new question rather than having to condition individually upon each possible response. Goldstein and Dillon's technique could result in different respondents obtaining at least a different sequence of questions and perhaps an entirely different set of questions. Since the order of presentation of questions

has been shown to make a statistically significant difference in questionnaire scores (cf. Payne [15], Oppenheim [16], and for an up-to-date bibliography and study, see Kalton, Collins, and Brook [17]), we desire a unified approach to variable selection not dependent upon the particular answers to previous questions. We select a single set and order of questions and give everyone the identical set in order.

Let x_{ijy} denote the number of respondents in group i ($i = 1, 2$) who answer j to question 1 and y to question Y , and $p(i, j, y)$ represent the corresponding proportion of respondents in group i with these answers. A test of the hypothesis that the inclusion of question Y yields no additional discriminatory power can be obtained by testing the hypothesis that the conditional distribution of Y is independent of the group classification given the answer to question 1, i.e.,

$$H_{1,Y}^{(0)}: p(iy|j) = p(i|j)p(y|j).$$

The worth of question Y is assessed by the p -value for rejecting $H_{1,Y}^{(0)}$. Using the directed divergence distance measure to test $H_{1,Y}^{(0)}$ yields the statistic

$$\begin{aligned} I(Y|1) &= 2I(p(iy|j) | p(i|j)p(y|j)) \\ &= 2 \sum_{i=1}^2 \sum_{j=1}^{k_1} \sum_{y=1}^{k_Y} x_{ijy} \ln \left(\frac{x_{ijy} x_{\cdot j \cdot}}{x_{ij \cdot} x_{\cdot j y}} \right) \end{aligned}$$

where the dot replacing a subscript is the usual convention for "sum over that variable", i.e., $x_{ij \cdot} = \sum_{y=1}^{k_Y} x_{ijy}$ and so on. The distribution of $I(Y|1)$ under $H_{1,Y}^{(0)}$ is asymptotically χ^2 with $k_1(k_Y - 1)$ degrees of freedom (cf. Kullback [2], or Gokhale and Kullback [12]). Hence if $p_Y = P[\chi^2(k_1, k_Y - k_1) \geq i(Y|1)]$ where $i(Y|1)$ is the observed value for $I(Y|1)$, we select the second question to minimize p_Y , $2 \leq Y \leq m$. The exact level of significance can be found from the distribution function $F(t) = 1 - (1 - t)^{m-1}$, $0 \leq t \leq 1$.

If the p -value for the second question is significant, we proceed on to select the third question by the same procedure. We test if the group classification and the answer to question Z , $3 \leq Z \leq m$ are conditionally independent given the response to questions 1 and 2. The information statistic is

$$2 \sum_{i=1}^2 \sum_{j=1}^{k_1} \sum_{k=1}^{k_2} \sum_{l=1}^{k_Z} x_{ijkl} \ln \left(\frac{x_{ijkl} x_{\cdot jk \cdot}}{x_{ij \cdot k} x_{\cdot jkl}} \right)$$

which is χ^2 with $k_1 k_2 (k_Z - 1)$ degrees of freedom. Again, the p -values for each question $3 \leq Z \leq m$ are compared to determine if the minimum p -value question is significant. If it is, we include it as question 3 and proceed until we obtain a nonsignificant result. When we finally find a nonsignificant result, we quit adding questions and consider the questionnaire complete.

One problem with utilizing contingency table methods of analysis is the rapid proliferation of cells, resulting in possible empty cells. Empty cells here will not bother us, but zero marginals will. The problem of zero marginals, and the resulting loss of degrees of freedom is discussed in

Gokhale and Kullback [12]. Another approach is to break the questionnaire into subsets of three or four questions each, each group of questions being treated separately. (These groups of questions are called "items" in the questionnaire literature.) For example, an item concerning economic status might be treated separately from an item concerning health, which in turn is treated separately from an item concerning education level. Each item is analyzed to obtain the questions in that particular set that should be included. The best subsets of each item are combined to form the overall questionnaire. This item analysis is preferred when there is item independence (cf. latent trait models—Hambleton and Cook [18]). Still another approach to the sparse cells problem is the "nearest neighbor" approach of Hills [19]. One groups together respondents whose previous answers differ in only one place on one question. This method is most useful when there is question dependence.

IV. DISCRIMINATION USING INFORMATION GAIN

The problem of discriminant analysis using categorical data has attracted much attention in the recent literature (cf. Lachenbruch [20], Goldstein and Dillon [11] and the bibliographies contained therein). If the number of variables is quite small, a contingency table approach is possible. For moderate numbers of variables, a log linear model may be fitted assuming certain higher order interaction terms vanish (cf. Gokhale and Kullback [12], and for computational techniques, see Brockett, Charnes, and Cooper [21]). The size needed here depends upon model complexity and data base size. If the questions all have ranked responses, or perhaps binary responses, then Fisher's linear discriminant function (LDF) (Fisher [22]) has proven to be quite effective for classifying respondents into their correct group (cf. Lachenbruch [20]). Fisher's LDF does not perform well, however, when the answers are not of a ranked character. Moore [23] refers to this as reversal in the likelihood ratio.

Discriminant analysis for discrete data involves two processes: first, one must score the categories, and second, one must combine the individual question scores to obtain an overall questionnaire score to be used for classification. The procedure we shall discuss here involves information theoretic scoring. This method is effective for nonordinal data, and outperforms Fisher's LDF when reversals are present.

Let $\mathbf{p}_i = (p_{i1}, \dots, p_{ik_i})$ and $\mathbf{q}_i = (q_{i1}, \dots, q_{ik_i})$ denote the group 1 and group 2 response probabilities for the k_i answers to question i . By scoring the i th answer via the information gain $\ln(p_{ii}/q_{ii})$ in favor of group 1 membership, we may transform the nonordinal data into ordinal data. The larger the score, the more likely is group 1 membership. (This is not true with raw scoring if the two groups' responses are polarized with respect to each other.) It should be noted that using information gain scoring coupled with classical LDF upon the transformed data has proven to be an effective technique for analysis of nonordinal categorical data (cf. Wolf and Brockett [24]).

TABLE I
PROBABILITY STRUCTURE OF THE SIMULATED QUESTIONNAIRE I

Question Number	Probability of Response of Group 1 to Answer					Probability of Response of Group 2 to Answer				
	I	II	III	IV	V	I	II	III	IV	V
1	25	10	30	10	25	10	35	10	35	10
2	35	10	10	10	35	10	35	10	35	10
3	35	10	10	10	35	10	25	30	25	10
4	25	10	30	10	25	10	25	30	25	10
5	25	25	25	10	15	15	10	25	25	25
6	15	25	25	25	10	10	25	25	25	15
7	25	25	30	10	10	10	10	30	25	25
8	10	35	35	10	10	10	10	35	35	10
9	35	10	35	10	10	10	10	35	10	35
10	35	35	10	10	10	10	10	10	35	35

PROBABILITY STRUCTURE OF THE SIMULATED QUESTIONNAIRE II

Question Number	Probabilities of Response of Group 1 to Answer					Probabilities of Group 2 to Answer				
	I	II	III	IV	V	I	II	III	IV	V
1	25	10	30	10	25	10	35	10	35	10
2	35	10	10	10	35	10	35	10	35	10
3	35	10	10	10	35	10	25	30	25	10
4	25	10	30	10	25	10	25	30	25	10
5	10	25	30	25	10	10	10	60	10	10
6	10	35	10	35	10	10	10	60	10	10
7	20	20	20	20	20	10	10	60	10	10
8	35	10	10	10	35	10	10	60	10	10
9	35	10	10	10	35	10	20	40	20	10
10	35	10	10	10	35	10	30	20	30	10

TABLE II
ERROR OF MISCLASSIFICATION IN PERCENT OF GROUP i
INTO GROUP j FOR 100 SAMPLES
FROM EACH GROUP $\epsilon(i, j)$

Questionnaire I		
Method	SPSS	DIG
Error (percent)		
$\epsilon(1, 2)$	21	1
$\epsilon(2, 1)$	17	3
Questionnaire II		
Method	SPSS	DIG
Error (percent)		
$\epsilon(1, 2)$	41	1
$\epsilon(2, 1)$	36	3

For simplicity we shall assume a first-order model of question interaction. In most applications to questionnaire analysis, the number of questions (variables) is fairly large and the number of respondents possible is constrained by time, money, availability, etc. For hierarchical contingency table models using log linear models, it has been found that for discrimination purposes, the first-order model outperforms the second- (or higher) order models even when dependence is present until the available sample size is roughly twice the number of possible states (cf. Chang, Lachenbruch, and Clarke [25]). Since we generally have an enormous number of possible cells in a large multiple

TABLE III
QUESTION WEIGHTS FOR THREE METHODS

Questionnaire I										
Question Method	1	2	3	4	5	6	7	8	9	10
Rao $F(1, 191)$	0.406	0.052	0.552	0.094	3.92	0	13.8	8	5.5	38.8
SPSS $F(1, 198)$	0.025	0.164	0.909	0.5	0.533	0.208	39.5	7.98	19.5	91.1
DIG $\chi^2(4)$	44.3	56.1	81.5	14.7	12.9	5	43.5	32	19.4	83.2
Questionnaire II										
Question Method	1	2	3	4	5	6	7	8	9	10
Rao $F(1, 194)$	0.182	0.134	6.444	0.168	0.058	0.356	0.002	0.001	0.76	2.49
SPSS $F(1, 198)$	0.0247	0.1643	2.909	0.4995	0.0396	0.0732	0.0035	0.0218	0.2262	1.1069
DIG $\chi^2(4)$	44.26	56.108	81.45	14.7	28.1	66.57	44.87	110.13	69.33	77.11

choice questionnaire, we usually have an insufficient number of respondents to justify a second-order model. Hence we shall assume a first-order model for each group.

Our assumption translates into the assumption of conditional independence of question responses given group membership. This is a common assumption in medical diagnostics (cf. Warner, *et al.*, [26], Bishop and Warner, [27], Boyle, *et al.*, [28], Nugent *et al.*, [29] or Reale, *et al.*, [30]), and is particularly reasonable in questionnaire analysis since the questions are usually designed so that the order of presentation does not matter. Thus a respondent answers question one, say on income, and then independently answers question two, say on history of heart disease, etc. It should be noted that this assumption of a first-order model does not imply lack of correlation between question responses for the observed overall group of respondents. Indeed, in practice there is much dependence between questions; however, this dependence is intimately connected to the group membership variable. This conditional independence assumption is also common in latent trait models for questionnaires (cf. Hambleton and Cook [18]).

If we are given samples of size n_1 and n_2 from group 1 and group 2, we may estimate p_{ti} and q_{ti} from these training samples, and pick a number s^* such that $\epsilon(1, 2) + \epsilon(2, 1)$ is minimized, where $\epsilon(1, 2)$ is the percentage of the n_1 respondents in group 1 with score $< s^*$, and $\epsilon(2, 1)$ is the percentage of the n_2 respondents in group 2 with score $> s^*$. We classify a respondent into group 1 if his questionnaire score is $\geq s^*$ and into group 2 otherwise; therefore $\epsilon(i, j)$ represents the percentage misclassified as belonging to group j . We note that the classification was not important in Section III, whereas here it is of primary importance. Our method will be called discrimination using information gain (DIG) for short.

Simulation studies were run to assess the power of this procedure relative to Fisher's LDF available in the statistical package for the social sciences (SPSS) computer

package. Also compared were the question weighting schemes of Rao [31], SPSS (cf. Cooley and Lohnes [32]), and the divergence weights from Section II.

The first two questionnaires we simulated are presented in Table I. Each question was simulated independently according to the given probability structure. Note that in Questionnaire I, questions 1, 2, 3 discriminate well but in a different way than do 5 and 10. The remaining questions in Questionnaire I discriminate moderately well except for 6, which essentially does not discriminate the groups.

Questionnaire II is a more extreme example of a questionnaire which discriminates very well, but upon which the SPSS Fisher method will produce distorted results since the centroids of the multivariate distributions generated by these probability structures coincide.

Table II gives the apparent error rates using Fisher's LDF as available in SPSS computer package and DIG. The percent of group i misclassified as belonging to group j is denoted $\epsilon(i, j)$. As predicted, DIG does much better for these types of questionnaires, since DIG does not depend upon the centroid separation of the two groups, but rather the "information gain" available from the questionnaire. This superiority of DIG also holds for the determination of significant questions. Table III shows the Rao, SPSS, and DIG weighting method. The Rao and SPSS methods distort the relative discriminatory power of the questions in Questionnaires I and II, whereas the DIG method correctly orders them according to the information present in the question for discrimination between the groups. The worth of a question which discriminates between the groups in a nonordinal way is assessed by DIG but not the other two. This is because the DIG procedure measures the "information distance" between the two probability measures corresponding to the two groups, and not the Euclidean distance between two real numbers (centroids) which ostensibly represent the groups. In fairness it should be noted that no claim of optimality for LDF in these situations has ever been asserted; however, due to easy availability of LDF on

computer packages, it is often used anyway. Hence, this comparison is useful.

We also ran the simulation for 10 000 samples in each group with Questionnaire I. For DIG, we found $\epsilon(1, 2) = 7.6$ percent, $\epsilon(2, 1) = 5.6$ percent, whereas for SPSS, $\epsilon(1, 2) = \epsilon(2, 1) = 20$ percent. With Questionnaire II using DIG we found $\epsilon(1, 2) = 5$ percent, $\epsilon(2, 1) = 4$ percent; for SPSS, $\epsilon(i, j) = 49$ percent. These results verify the theoretical conclusion that for this type of questionnaire, with discriminating but nonordinal questions, DIG is preferred. Examples of such questions might be marital status, or questions on political extremes. The DIG weights are also more reliable indicators of question discriminatory power.

V. APPLICATIONS

We shall present several examples in which the information theoretic methods of analysis were used on actual data. We consider first a psychiatric screening questionnaire developed by Dr. H. Davidian, Head of the Department of Psychiatry, University of Tehran, Tehran, Iran. In this questionnaire the permitted responses to questions such as "Do you feel restless?" are "never", "occasionally", "frequently", "always". The questionnaire was designed so that each question measures the degree of some aspect of the respondent's "mental stress". The question responses are all ordered by degree in the same direction; "mentally ill" patients are presumed to respond to the "high" end, "normals" at the "low" end. Since this is so, the two groups' score centroids should be well separated and consequently, both LDF and DIG should discriminate well.

The questionnaire consisted of 46 questions and was designed for the purpose of classifying each respondent as "mentally ill" or "normal". It was given to 143 respondents, 90 of whom were classified prior to the administration of the questionnaire as "mentally ill" while the rest were "normal". The values assigned to the responses were 0 for "never", 1 for "occasionally", 2 for "frequently" and 3 for "always". SPSS was run using this raw scoring technique. Essentially SPSS and DIG behave the same for this nice ordinal data. The question weights developed by SPSS and DIG, as expected, gave essentially the same assessment of question worth. The question weights were used to shorten the questionnaire. The 22 questions with the highest weights were selected. (Psychiatric technical considerations also played a part in the choice of these questions.) Using this new "reduced" questionnaire it was found that the apparent error rate was $\epsilon(1, 2) = \epsilon(2, 1) = 0$ for the DIG analysis, whereas for the SPSS analysis it was $\epsilon(1, 2) = 0$ and $\epsilon(2, 1) = 4$ percent. This questionnaire has been used for screening purposes in Iran. The use of the reduced questionnaire has resulted in a considerable saving in time over the original questionnaire. (Note: in all of these simulations the apparent error rate is used, and hence is optimistically biased.)

The second set of data upon which these methods have been used involved a survey conducted by the Pan Ameri-

TABLE IV
SIMULATION OF RESPONSE PROBABILITIES FOR PAN AMERICAN
HEALTH ORGANIZATION SURVEY

Question Number	Probability of Response of Group 1 to Answer					Probability of Response of Group 2 to Answer				
	I	II	III	IV	V	I	II	III	IV	V
1	17	37	46	0	0	0	29	66	5	0
2	76	22	2	0	0	55	50	5	0	0
3	69	25	6	0	0	47	45	8	0	0
4	93	7	0	0	0	87	13	0	0	0
5	19	16	65	0	0	10	24	66	0	0
6	39	15	23	23	0	32	18	16	34	0
7	24	15	22	26	13	11	34	21	18	16
8	92	8	0	0	0	99	1	0	0	0
9	32	17	14	12	25	16	28	19	13	24
10	70	1	29	0	0	53	0	47	0	0
11	68	29	3	0	0	45	50	5	0	0
12	63	37	0	0	0	76	24	0	0	0

TABLE V
ERROR IN PERCENT OF MISCLASSIFICATION: QUESTIONNAIRE III
(100 SAMPLES IN EACH GROUP)

Method Error	SPSS	DIG
$\epsilon(1, 2)$	26	12
$\epsilon(2, 1)$	25	24

can Health Organization (PAHO) on child mortality in 1969-1970 in South American countries. Among the questionnaires was one covering the socioeconomic status of a household and various environmental factors. Some questions were "What type of water supply do you have?" with such permitted responses as "piped water", "well", "rain water"; and "What is the marital status of the mother?" with the responses "married", "divorced", "separated". We note here that many of these questions had answers that were not essentially ordinal; hence, reversals may occur, and a linear discriminant function may be inappropriate. (Because of a contractual agreement between PAHO and the World Health Organization, (WHO), Geneva, Switzerland, where these data were analyzed, we are not able to disclose details of this questionnaire or of the analyses.) We analyzed 12 questions from this questionnaire. Group 1 consisted of all those households where a child under five years died of malnutrition or diarrhea. The second group consisted of households where a child died from other causes. In all there were 952 households; however, complete information was available only on 154 in group 1, and 37 in group 2. We chose only those questionnaires without missing responses since SPSS needs a special program for missing data (DIG does not), and we wanted a direct comparison between the two methods. The proportion of answers of each group to each response is given in Table IV. From Table V we find that for DIG, $\epsilon(1, 2) = 12$, $\epsilon(2, 1) = 24$, and for SPSS, $\epsilon(1, 2) = 26$, $\epsilon(2, 1) = 25$. It should be noted that the results of SPSS

TABLE VI
QUESTION WEIGHTS: QUESTIONNAIRE III

Question Method	1	2	3	4	5	6	7	8	9	10	11	12
Rao	12.6	4.48	0	4.9	1.89	0.48	0.128	1.39	0.808	2.41	11.5	0.09
SPSS	24.9	10.6	0.966	7.96	2.82	0.929	0.07	4.78	0.106	3.95	27.7	2.82
DIG	34.3	10.8	4.85	8.45	4.85	4.45	8.11	5.52	11.2	3.95	28.1	2.82

and DIG agree upon which of the questions are significantly discriminating (Table VI) except for question 9, which DIG found to be highly discriminating and SPSS found to be not discriminating. The importance of question 9 was lost to SPSS since it discriminated in a nonordinal manner. Consequently, if one uses SPSS on data that is not essentially ordinal, one may unintentionally eliminate significant variables.

As a final example of how this method has been used we briefly sketch the following. The acceptability group in the human reproduction division of the WHO has used the DIG technique to assess the feasibility and acceptability of various modes of contraception. In particular they used this method to determine for which groups of people a paper birth control pill is acceptable. Various factors affect the acceptability of the paper pill. For example, the persons involved may refuse to eat paper, the climate may be such that the paper cannot be kept dry, or the life style may be such that the paper sheets cannot be kept clean. On the other hand if the paper pill is acceptable in a particular region, it reduces costs and is easier to store and administer. A categorical questionnaire was designed to ascertain which groups would accept the paper pill and was given to samples in Alexandria and Cairo, Egypt; Cariche and Ibadan, India; Manila, the Philippines; Stockholm, Sweden; and Bangkok, Thailand. It was desired to find which questions or factors discriminated between those respondents who accepted the paper pill and those who did not. Also it was desirable to know how effective each question was in distinguishing between the groups.

Both SPSS and DIG analysis was performed on the data using sample sizes of about 200 in each country. Both produced the same set of discriminating questions, and approximately the same error rates (20–30 percent). We cannot present a detailed description of the data collected since this study is still ongoing, and the WHO has priority on the publication of the exact data. Nevertheless, this example and previous examples show how information theory has been successfully applied to real data. For further information on the contraceptive study, contact Dr. B. Cristian, Human Reproduction Division, WHO, Geneva, Switzerland. There is also a computer program written by Busca and Diethelm at WHO that can be used to implement the DIG analysis.

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Schur Recursions, Error Formulas, and Convergence of Rational Estimators for Stationary Stochastic Sequences

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Abstract—An exact and approximate realization theory for estimation and model filters of second-order stationary stochastic sequences is presented. The properties of J -lossless matrices as a unifying framework are used. Necessary and sufficient conditions for the exact realization of an estimation filter and a model filter as a submatrix of a J -lossless system are deduced. An extension of the so-called Schur algorithm yields an approximate J -lossless realization based on partial past information about the process. The geometric properties of such partial realizations and their convergence are studied. Finally, connections with the Nevanlinna-Pick problem are made, and how the techniques presented constitute a generalization of many aspects of the Levinson-Szegő theory of partial realizations is shown. As a consequence generalized recursive formulas for reproducing kernels and Christoffel-Darboux formulas are obtained. In this paper the scalar case is considered. The matrix case will be considered in a separate publication.

I. INTRODUCTION

LET $x(t)$, $t = 0, \pm 1, \dots$, denote a real stationary zero mean scalar stochastic sequence with covariance function

$$r(k) = E[x(t)x(t-k)] = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-ik\theta} W(e^{i\theta}) d\theta$$

and spectral density W .

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A shaping (or modeling) filter for this stochastic process is a linear time-invariant causal filter which when driven by white noise produces an output with covariance equal to $r(k)$. An innovations or estimation filter is the inverse of a shaping filter. It produces white noise when it is driven by the given stochastic sequence. In practice it is of interest to establish *recursive procedures* for computing approximate shaping and innovations filters and thus avoid having to recompute the whole filter each time the permitted complexity is increased: the additional improvement should be achieved by adding sections without changing the existing structure. The same philosophy underlies both the "ladder structure" used in modern digital filter design and the theory of orthonormal expansions: the initial coefficients $f_j = (f, \phi_j)$ of f with respect to an orthonormal set of functions ϕ_j , $j = 1, \dots, n$, in a Hilbert space with inner product $(\cdot, \cdot)$ do not have to be recomputed as the size of the set changes.

Levinson [1] established a recursive procedure for computing a sequence of approximate autoregressive (AR) shaping filters from the covariance data which, under mild conditions, converge to the exact shaping filter. The theory of these filters is intimately connected with the theory of polynomials on the unit circle which are orthogonal with respect to the spectral density of the underlying stochastic sequence. In the paper of Dewilde, Vieira, and Kailath [2] it was shown that the Levinson procedure could be viewed as a special case of an algorithm for coprime factorization. In this paper we exploit the greater flexibility afforded by